SCIENTIFIC COMPUTATION
COMPSCI 3X03/SFWRENG 4X03

NED NEDIALKOV
DEPARTMENT OF COMPUTING AND SOFTWARE

MCMASTER UNIVERSITY
FALL 2025

Lectures:	Monday, Thursday	1:30PM–2:20PM	ITB	137
	Wednesday	1:30PM–2:20PM	HSC	1A1
Tutorials:	Wednesday	8:30AM–9:20AM	T13	105
	Wednesday	2:30PM–3:20PM	T13	106
	Wednesday	2:30PM–3:20PM	BSB	117
	Friday	12:30PM–1:20PM	T13	106
Office hours:	Monday	2:30PM–3:30PM	ITB	123

Instructor
Ned Nedialkov, ITB 123

Teaching assistants
Amin Toloekashefpakdel
Hrad Ghoukasian
Kyanna Dagenais
Nicholas Petrunti
Zhaleh Rahimi

Course Overview

This is an introductory course in scientific computing. We will study issues related to floating-point arithmetic in numerical computing and explore a variety of numerical algorithms. Topics include numerical differentiation and integration, solving linear and nonlinear systems, eigenvalues, least-squares problems, and numerical methods for ordinary differential equations. The course will also provide a brief introduction to deep learning. We will emphasize error analysis, stability, and convergence of numerical algorithms. All programming will be carried out using MATLAB as both the language and computing environment.

Calendar description.

Computer arithmetic and roundoff error analysis. Interpolation, integration, solving systems of linear and nonlinear equations. Eigenvalues and singular value decomposition. Numerical methods for ordinary differential equations.

Three lectures, one tutorial (one hour); first term

SFWRENG 4X03
Prerequisite(s): Both MATH 1ZB3 and MATH 1ZC3, or both MATH 1AA3 and 1B03; SFWRENG 2C03; registration in any Software Engineering program
Antirequisite(s): CHEMENG 2E04, CIVENG 2E03, COMPENG 3SK3, 3SK4, COMPSCI 3X03, 4X03, ENGPHYS 2CE4, 3NM4, MATH 3NA3, 4NA3, MECHENG 3F04, SFWRENG 3X03, MECHTRON 3X03

COMPSCI 3X03
Prerequisite(s): MATH 1AA3 or 1ZB3, and MATH 1B03 or 1ZC3; COMPSCI 2C03; One of MATH 2C03 or 2Z03 is recommended.
Antirequisite(s): COMPENG 3SK3, 3SK4, COMPSCI 4X03, SFWRENG 3X03, 4X03. MECHTRON 3X03, MATH 3NA3, 4NA3, CHEMENG 2E04, CIVENG 2E03, ENGPHYS 2CE4, 3NM4, MECHENG 3F04. Not open to students in a Software Engineering or Mechatronics Engineering program

LEARNING OBJECTIVES

Postcondition. A *learning objective* for a course is something the student is expected to know and understand or to be able to do by the end of the course. The learning objectives for this course are given below. Taken together, this set of learning objectives constitute the *postcondition* of the course.

1. Students should know and understand
   (a) issues in floating-point (FP) computations, overflow, cancellations, roundoff errors
   (b) truncation errors
   (c) solving linear systems
   (d) linear least squares
   (e) solving nonlinear equations
   (f) interpolation
   (g) numerical integration
   (h) stability of methods for initial-value problems in ordinary differential equations
2. Students should be able to
   (a) perform roundoff error analysis
   (b) derive error bounds
   (c) interpret numerical results
   (d) perform simple complexity analysis
   (e) analyze convergence
   (f) analyze stability
   (g) write Matlab programs implementing numerical methods

Precondition The *precondition* of the course is the set of university-level learning objectives that the student is expected to have achieved before the start of the course. The precondition includes knowledge of calculus and linear algebra

MATERIALS AND FEES

Required:

- Numerical Mathematics and Computing (7th Edition), Retail Price: $283.95
- *eBook: Numerical Mathematics and Computing*, Retail Price: $77.95

Optional: Student Solutions Manual, Retail Price: $97.95

Calculator: McMaster standard calculator required for midterm test and exam.

Note: The above prices are from the official textbook site. The 6th edition of the textbook is fine; however, there are some mismatches between the 7th and 6th edition in the numbering of sections, exercises, etc.

SCHEDULE OF TOPICS (TENTATIVE)

These may be updated as we progress with the course.
§X: Y means Section X should be studied and exercise Y is suggested
The corresponding numbers from the 6th edition are given in $(\cdots)$.

Week 1, 2: Introduction, floating-point arithmetic

- Chapter 1
  - §**1.1**: 8, 10 (§**1.1**: 8, 10)
  - §**1.2**: 8, 9, 23, 30, 41 (§**1.2**: 8, 9, 23, 30, 41)
  - §**1.3**: 13, 14, 18, 22, 24, 45 (§**2.1**: 13, 14, 18, 24, 45)
  - §**1.4**: 8, 11, 13, 14, 24, 26, 29 (§**2.2**: 8, 11, 14, 24, 26, 29)
    Note: you can skip the theorem on loss of precision

Week 3, 4: Linear systems

- Chapter 2
  - §**2.1**: 1, 2, 7a (§**7.1**: 1, 2, 7a)
  - §**2.2**: 1, 8, 9, 23 (§**7.2**: 1, 8, 9, 23)
- Chapter 8

- **§8.1, pp. 358–365, 371–373**: 2, 4, 11, 15, 18, 23 (**§8.1, pp. 293–302, 306–307**: 2, 4, 11, 15, 18, 23)

Week 5–6: Interpolation

- Chapter 4
  - **§4.1, pp. 153–168**: 1, 9, 12, 18, 40 (**§4.1, pp. 124–141**: 1, 9, 12, 18, 40)
  - **§4.2**: 5, 9, 10, 15 (**§4.2**: 5, 9, 10, 15)

Week 7, 8: Numerical integration

- Chapter 5
  - **§5.1, up to p. 210**: 1, 6, 7, 8, 11 (**§5.2, up to p. 196**: 2, 4, 5, 7, 19)
  - **§5.3**: 1, 2, 4 (**§6.1**: 1, 2, 4)

Week 9: Linear least-squares

- Chapter 9
  - **§9.1**: 3, 5, 13, 25 (**§12.1**: 3, 5, 13, 25)

Week 10: Solving nonlinear equations

- Chapter 3
  - **§3.2**: 1, 3, 13, 14, 23 (**§3.2**: 1, 3, 13, 14, 23)
  - **§3.1**: 8, 12, 13 (**§3.1**: 8, 12, 13)
  - **§3.3**: 2, 3 (**§3.3**: 2, 3)

Week 11: Introduction to deep learning

Week 12: Initial-value problems

- Chapter 7

Assessments and Schedule

Evaluation

- Assignments: $4 \times 5\% = 20\%$
- Midterm: 25%
- Final exam: 55%

The final exam will cover the whole course material. The date of the final examination will be set by the Registrar. Students must be available for the entire examination period.

Exam aids. For both the midterm and the final exam, you may bring one sheet of paper (8.5” $\times$ 11”). You may use both sides of the sheet. The sheet may contain any course material (formulas, notes, worked examples, etc.), prepared either in your own handwriting or typed, and in any font size.

Schedule. These are tentative dates and may change depending on how we progress with the material.

Component	Date
Assignment 1	22 September – 6 October
Assignment 2	6 October – 23 October
Midterm	27 October, during class time
Assignment 3	3 November – 17 November
Assignment 4	17 November – 1 December

- Assignments will be posted on Avenue to Learn.
- All submissions must be made through Avenue to Learn.
- Graded work will normally be returned within about 10 days of submission.

Course Policy

Lectures

- All course material will be posted on Avenue.

Assignments

- Late work will not be graded without an MSAF.

Missed work

- The MSAF accommodation for a missed assignment is a **three calendar day** extension from the original assignment deadline.
- The MSAF accommodation for a missed midterm is to roll the weight of the midterm into the weight of the final examination.

Use of Artificial Intelligence (AI) Tools. Students are not permitted to use generative AI in this course. In alignment with McMaster academic integrity policy, it "shall be an offence knowingly to … submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also stated in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Changes

- The instructor reserves the right to modify elements of this course and will notify students of any changes through Avenue to Learn.

Resources

- MATLAB Onramp
- Get Started with MATLAB
- Floating Points
- D. Goldberg. What Every Computer Scientist Should Know About Floating-Point Arithmetic

Appendix. Approved Advisory Statements

Academic Integrity. You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy.

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism Detection. Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g. online search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to `http://www.mcmaster.ca/academicintegrity`.

Courses with an Online Element. Some courses may use online elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses online elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring. Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock or restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations. As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the *Code of Student Rights & Responsibilities* (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. Avenue 2 Learn, WebEx, Zoom) will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Academic Accommodation of Students With Disabilities. Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or `sas@mcmaster.ca` to make arrangements with a Program Coordinator. For further information, consult McMaster University's *Academic Accommodation of Students with Disabilities* policy.

Requests For Relief For Missed Academic Term Work. In the event of an absence for medical or other reasons, students should review and follow the *McMaster Student Absence Form Policy (MSAF Policy)*.

Academic Accommodation For Religious, Indigenous Or Spiritual Observances (RISO). Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the *RISO policy*. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording. Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances. The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g. severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn, and/or by McMaster email.